

COMP 3311: Database Management Systems

Tutorial 10 Query Processing

Exercise 1: Given the following relations and the information about them, process the relational algebra tree for the query on the right using a pipelined plan and answer the following questions.

Student(studentId, sName, gender)
1,000 tuples; 100 pages; index on studentId

EnrollsIn(studentId, courseId, year)
6,000 tuples; 600 pages; index on courseId

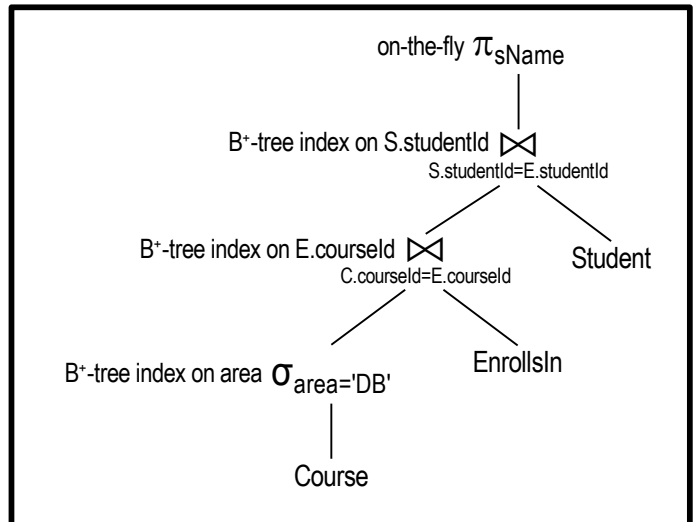
Course(courseId, cName, area, credit)
200 tuples; 40 pages; index on area;
10 different areas, with 20 tuples per area

All foreign keys are not null.

All indexes are B⁺-tree clustering indexes with 4 levels.

EnrollsIn tuples are *uniformly distributed* among students and courses.

a) Estimate the query result size.



```
select sName
from Student S, EnrollsIn E, Course C
where S.studentId=E.studentId
and C.courseId=E.courseId
and area= 'DB';
```

b) Estimate the query page I/O cost using a pipelined plan (i.e., do not store intermediate results).

Step 1: $\sigma_{\text{area}='DB'}$ Course $\Rightarrow$ result A

Strategy:

Strategy explanation and cost calculation

Step 1 page I/O cost: _____

Step 2: result A $\bowtie$ EnrollsIn $\Rightarrow$ result B

Strategy:

Strategy explanation and cost calculation

Step 2 page I/O cost: _____

Step 3: result B $\bowtie$ Student

Strategy:

Strategy explanation and cost calculation

Step 3 page I/O cost: _____

Query processing page I/O cost: _____

Name: (1) _____ / _____ Student#: (1) _____ Date: _____
Family/Given (PRINT) Given/First (PRINT)

Name: (2) _____ / _____ Student#: (2) _____
Family/Given (PRINT) Given/First (PRINT)

This is a partner exercise. YOU MUST DO THIS EXERCISE WITH A PARTNER. Individual submissions will not be graded.
A genuine effort must be made to complete all exercises to obtain full marks.
SUBMIT ONLY ONE COMPLETED PARTNER WORKSHEET.

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Exercise 2: For the query $\pi_{A,B,C,D}(R \bowtie_{A=C} S)$ assume that the projection routine uses external sorting, eliminates all unwanted attributes during the initial sort pass and removes duplicate tuples on-the-fly during the merge passes. Furthermore, assume the following:

- R is 10 pages; each R tuple is 300 bytes.
- The combined size of attributes A, B, C and D is 450 bytes.
- A is a key for R.
- The page size is 1024 bytes.
- Only the optimized block nested-loop join method is implemented.
- S is 100 pages; each S tuple is 500 bytes.
- A and B are in R and have combined size of 200 bytes; C and D are in S
- Each S tuple joins with exactly one R tuple.
- The buffer size M is 3 pages.

a) Join Cost (using optimized block nested-loop and eliminating unwanted attributes during the join)

Join page I/O cost: _____

Write join result page I/O cost: _____

b) Projection Cost (using external sorting)

Sort page I/O cost pass 0: _____

Merge page I/O cost: _____

Query processing page I/O cost: _____

Name: (1) _____ / _____ Student#: (1) _____ Date: _____
Family/Given (PRINT) Given/First (PRINT)

Name: (2) _____ / _____ Student#: (2) _____
Family/Given (PRINT) Given/First (PRINT)

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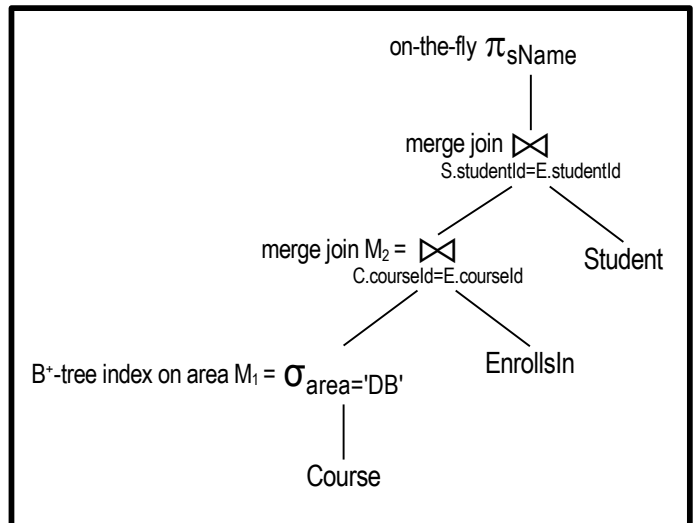
Exercise 3: For the query of Exercise 1, suppose the intermediate results $M_1 = \sigma_{\text{area}='DB'} \text{Course}$ and $M_2 = M_1 \bowtie_{\text{courseId}} \text{Enrollsn}$ are materialized (i.e., written to disk) for the relational algebra tree shown on the right. What is the query processing page I/O cost for this relational algebra tree if merge join is used for all joins?

Assume there are 22 buffer pages and *attributes in the same relation all have the same size*.

Step 1: Cost to materialize $M_1 = \sigma_{\text{area}='DB'} \text{Course}$

Strategy: index lookup using B⁺-tree on area

Strategy cost calculation



Step 1 page I/O cost to materialize M_1 : _____

Step 2: Cost to materialize $M_2 = M_1 \bowtie_{\text{courseId}} \text{Enrollsn}$

Strategy: merge join

Strategy cost calculation

Step 2: Cost to materialize $M_2 = M_1 \bowtie_{\text{courseId}} \text{EnrollsIn}$

Strategy: merge join

Strategy cost calculation

Step 2 page I/O cost to materialize M_2 : _____

Step 3: Cost to compute $M_2 \bowtie_{\text{studentId}} \text{Student}$

Strategy: merge join

Strategy cost calculation

Step 3 page I/O cost: _____

Query processing page I/O cost: _____